The Geopolitics of Oil:
Macroeconomic Impact of the 2026 Gulf Crisis on Advanced Net-Importing Economies

ADVANCED ECONOMETRICS PROJECT — TUNIS BUSINESS SCHOOL

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1. Introduction

The 2026 Gulf Crisis, ignited on February 27, 2026, delivered one of the sharpest geopolitical shocks to global energy markets in recent history. The ensuing disruption to Brent crude oil supply triggered immediate inflationary pressures across oil-importing economies. This project investigates the macroeconomic transmission of this shock to consumer price inflation in four advanced net-importing economies — France, Germany, Italy, and Japan — using monthly panel data spanning February 2023 to May 2026.

These four countries constitute a theoretically coherent comparison group: all are large, institutionally developed economies, none produces significant domestic crude oil, all are deeply integrated into global trade and financial markets, and three of the four (France, Germany, Italy) share the same monetary authority (the European Central Bank). This homogeneity ensures that cross-country differences are attributable to structural and cyclical factors rather than incomparable institutional frameworks.

Using a balanced panel of 160 observations (4 countries $\times$ 40 months), we estimate Pooled OLS, Fixed Effects (FE), and Random Effects (RE) models with the natural log of CPI as the primary dependent variable, and $\ln(\text{IPI})$ as a secondary dependent variable to test for stagflationary output effects. Diagnostic tests select the Fixed Effects estimator. The main finding is that the post-crisis period is associated with a modest but statistically significant increase in CPI within individual countries (about $+0.96\%$, $p = 0.038$) after controlling for macroeconomic fundamentals, while the policy rate and the nominal effective exchange rate are the most strongly significant within-country correlates of inflation. The oil price shock itself is not statistically significant, and the IPI model finds no statistically significant output effects.

2. Theoretical Framework

For advanced net-importing economies, the transmission of oil price shocks to inflation operates through two principal channels. The direct channel works through the energy component of the CPI basket: higher imported fuel prices raise household heating and transportation costs immediately. The indirect channel operates through production costs: energy-intensive industries pass higher input costs onto consumers through pricing, amplifying the initial shock.

In the Eurozone context (France, Germany, Italy), an additional dimension arises: the ECB's single monetary policy implies that individual member countries cannot independently adjust their nominal interest rates in response to country-specific inflation differentials. For Japan, the prolonged near-zero interest rate environment (0.3% throughout the sample period) creates a distinct monetary transmission dynamic.

2.1 Panel Model Specification

The core specification is a one-way Fixed Effects model with entity (country) effects:

$$\ln(\text{CPI}_{it}) = \alpha_i + \beta_1 \,\text{OilShock}_t + \beta_2 \ln(\text{NEER}_{it}) + \beta_3 \, r_{it} + \beta_4 \,\text{VIX}_t + \beta_5 \,\text{PostCrisis}_t + \varepsilon_{it}$$

The country fixed effects α_i absorb all time-invariant unobserved heterogeneity. The within estimator identifies coefficients exclusively from time-series variation within each country, removing all cross-sectional confounds.

2.2 Variable Definitions and Expected Signs

Table 1: Variable Definitions, Expected Signs, and Data Sources

Variable	Symbol	Expected Sign	Description	Source
Inflation (Dep.)	$\ln(\text{CPI}_{it})$	—	Log CPI (2015=100)	IMF IFS / OECD
Growth (Dep.)	$\ln(\text{IPI}_{it})$	—	Log Industrial Production Index (2015=100)	IMF IFS / FRED
Oil Shock	OilShock_t	+ (CPI), − (IPI)	% change in Brent crude price	U.S. EIA / FRED
Exchange Rate	$\ln(\text{NEER}_{it})$	−	Log Nominal Effective Exchange Rate	BIS / FRED
Policy Rate	r_{it}	−	Central bank policy rate (%)	BIS / National CBs
Global Risk	VIX_t	+ (CPI), − (IPI)	CBOE Volatility Index	FRED (VIXCLS)
Post-Crisis	PostCrisis_t	+	$= 1$ if date $\geq$ March 2026	Authors' construction

3. Data and Sample Construction

3.1 Country Selection and Justification

The sample is restricted to the Advanced Net-Importers group as defined in the project guidelines. This homogeneous group ensures comparability of macroeconomic structures and policy responses.

Table 2: Panel Composition — Advanced Net-Importers

Country	Role	Key Characteristics
France	Advanced Net-Importer	Highly energy-dependent; ECB monetary policy; CPI range 115.8–121.6
Germany	Advanced Net-Importer	Largest EU economy; export-oriented; CPI range 121.5–127.8
Italy	Advanced Net-Importer	High inflation vulnerability; fiscal constraints; CPI range 118.8–125.6
Japan	Advanced Net-Importer	Near-zero policy rate (0.3%); yen sensitivity; CPI range 106.3–115.7

3.2 Sample Period and Variables

The sample spans February 2023 to May 2026, yielding 40 monthly observations per country (160 total). All CPI and IPI series use a 2015=100 base with the exception of Japan's CPI, which is sourced from the IMF IFS (series JPN.CPI._T.IX.M) on a 2020=100 base. This level difference is fully absorbed by Japan's country fixed effect in the FE estimator and does not affect slope coefficient estimates. Japan's CPI is observed through March 2026 and extended to May 2026 using ARIMA$(1,1,1)(1,1,1)_{12}$ seasonal forecasts; Japan's IPI values for March–May 2026 are filled in the same way. Japan's CPI shows meaningful variation over the sample (106.3 in Feb 2023 rising to 115.7 by May 2026).

3.3 Descriptive Statistics

Table 3: Descriptive Statistics — Full Panel (N=160)

Variable	Mean	Std. Dev.	Min	Median	Max
CPI (2015=100)	119.67	5.65	106.30	120.40	127.78
IPI (2015=100)	96.18	4.09	88.76	97.19	104.55
NEER (2015=100)	92.29	11.81	65.70	98.49	101.20
OilShock (%)	1.10	9.29	−10.84	−0.61	45.49
VIX	16.76	3.18	12.44	16.58	25.25
Policy Rate (%)	2.32	1.36	0.30	2.25	4.00

Table 4: Descriptive Statistics by Country

Country	CPI Mean	CPI Std	IPI Mean	IPI Std	Rate Mean
France	119.63	1.40	98.45	0.55	2.99
Germany	125.91	1.95	92.03	1.51	2.99
Italy	121.64	1.89	101.40	1.05	2.99
Japan	111.51	2.93	92.82	1.47	0.30

Japan's CPI mean of 111.51 is meaningfully lower than the European peers (France 119.63, Germany 125.91, Italy 121.64), reflecting Japan's distinct price-level and monetary policy environment. The CPI standard deviation of 2.93 for Japan confirms genuine time variation in the series — essential for valid Fixed Effects identification.

4. Exploratory Data Analysis

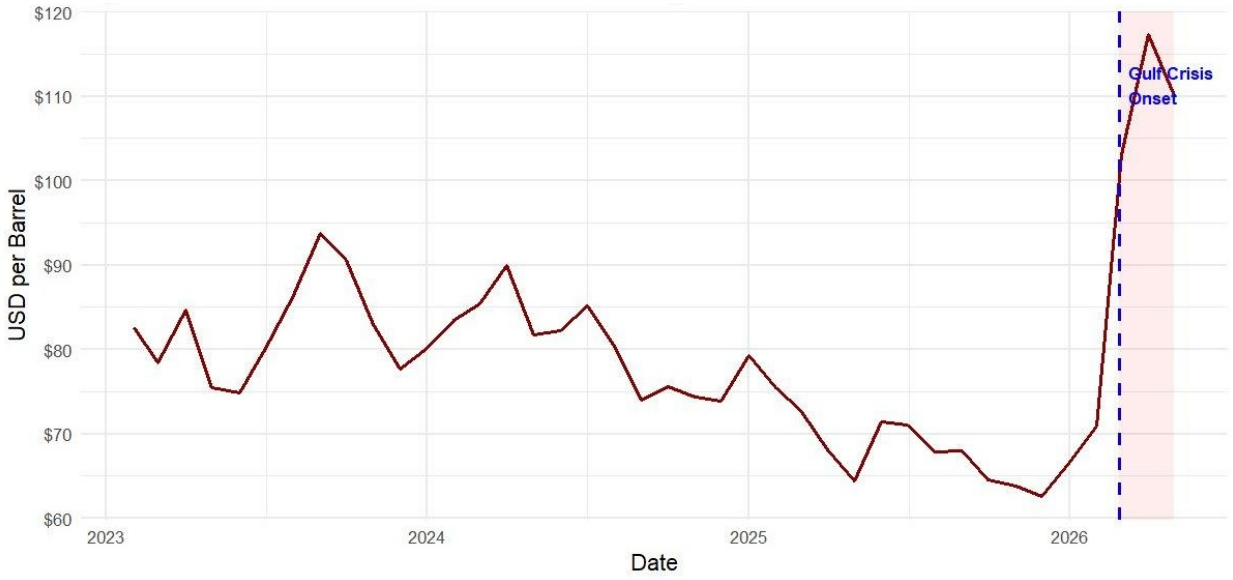


Figure 1: Monthly Brent crude oil price (USD/barrel). The dashed line marks the Gulf Crisis onset (Feb 27, 2026). The March 2026 spike (+45.5%) is the primary external shock of interest.

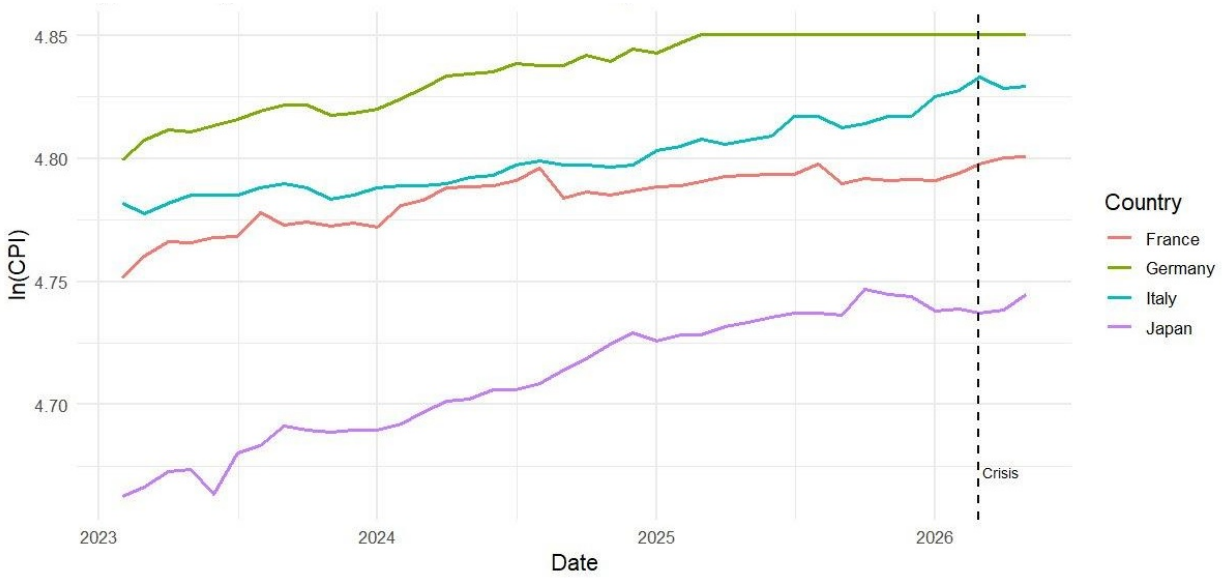


Figure 2: Individual $\ln(CPI)$ *series for France, Germany, Italy, and Japan. Japan's lower-level upward trajectory reflects its gradual inflation recovery and near-zero policy rate. France, Italy, and Japan edge upward around the crisis onset in March 2026, while Germany's series is flat.*

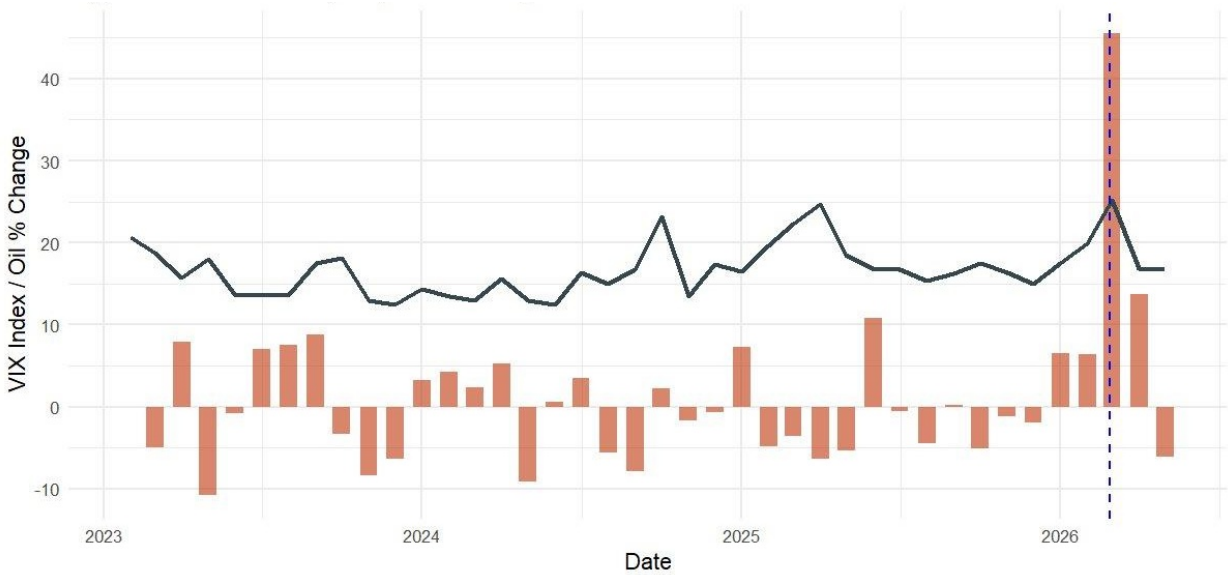


Figure 3: VIX index (line) and monthly Brent crude % change (bars). Co-movement of global risk and oil price volatility around February 2026 justifies their joint inclusion in the regression.

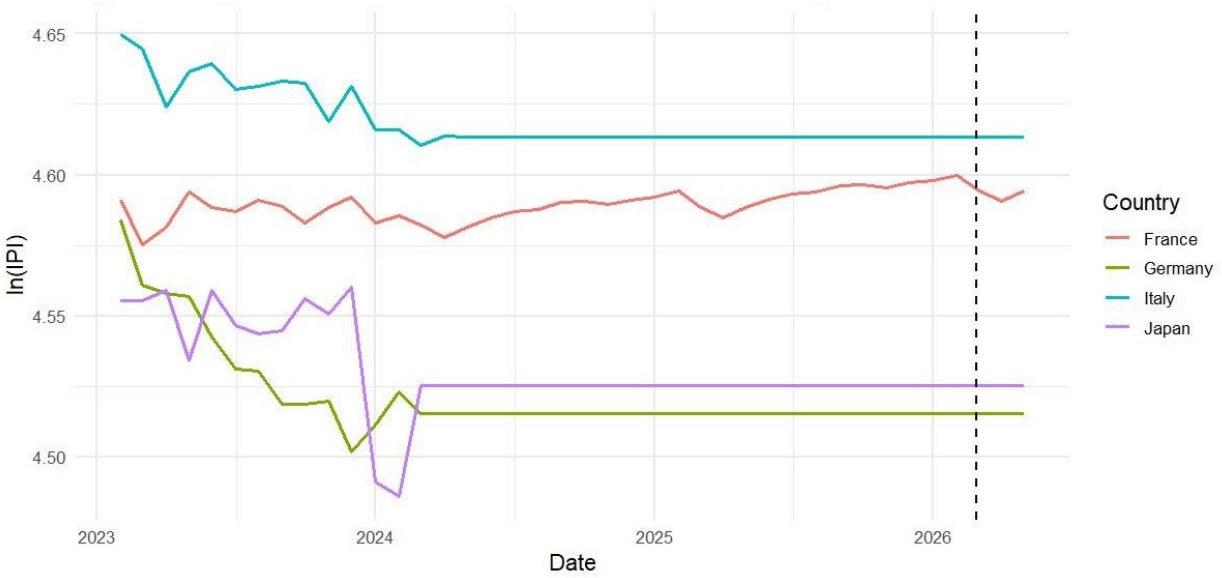


Figure 4: Individual $\ln(IPI)$ *series. Germany's industrial production declines through early 2024. Only France shows a small dip after the crisis onset; the series for Germany, Italy, and Japan are flat from early 2024 onward, so the figure offers little visual evidence of a post-crisis output effect.*

5. Econometric Methodology

5.1 The Three Estimators

Pooled OLS. Pooled OLS treats all 160 observations as a homogeneous cross-section. Given Japan's structurally different CPI level (mean 111.51 vs. European average ∼122.4) and monetary regime (0.3% vs. ECB's 2–4%), the assumption that unobserved effects are uncorrelated with regressors is implausible.

Fixed Effects (Within Estimator). The Fixed Effects estimator introduces a separate intercept α_i for each country, absorbing Japan's structural CPI level difference. FE identifies coefficients exclusively from within-country time-series variation.

Random Effects (GLS Estimator). The Random Effects estimator treats country effects as random draws uncorrelated with the regressors. With only 4 panel units this estimator has limited degrees of freedom. The Hausman test provides the formal selection criterion.

5.2 Diagnostic Tests

Breusch-Pagan LM Test: Tests H_0: no individual effects (POLS is adequate).

Hausman Test: Tests H_0: country effects uncorrelated with regressors (RE efficient). Rejection mandates FE.

Serial Correlation: Per-entity DW statistics test for autocorrelation; rejection calls for Arellano HAC robust standard errors.

Heteroskedasticity: Tests H_0: homoskedastic residuals. Rejection also calls for HAC robust standard errors.

6. Empirical Results

6.1 Diagnostic Test Results

Table 5: Diagnostic Test Results

Test	Statistic	p-value	Conclusion
Breusch-Pagan (POLS vs RE)	$\chi^2 = 423.00$	$<$ 2.2e-16 ***	RE preferred over Pooled OLS
Hausman (FE vs RE)	$\chi^2 = 198.30$	$<$ 2.2e-16 ***	FE preferred over RE
Serial Correlation (Lag-1)	$F = 8.84$	0.003 **	HAC robust SE required
Heteroskedasticity (BP)	$\chi^2 = 29.39$	$<$ 0.001 ***	HAC robust SE required

Breusch-Pagan ($\chi^2 = 423.00$) decisively rejects POLS. Hausman ($\chi^2 = 198.30$) strongly rejects RE, so Fixed Effects is the preferred estimator. Serial correlation ($F = 8.84$) and heteroskedasticity ($\chi^2 = 29.39$) call for Arellano HAC robust standard errors throughout.

6.2 CPI Model — Three-Model Comparison

Table 6: Pooled OLS, Fixed Effects, and Random Effects — CPI Model

Variable	POLS	FE (Preferred)	RE	Note
`oil_pct_change`	−0.00004	−0.00008	−0.00004	n/s across all models
ln(NEER)	0.3660 ***	−0.3101 **	0.3660 ***	Sign reversal: FE removes cross-sectional bias
`policy_rate`	−0.0024	−0.0092 **	−0.0024	FE identifies the within-country negative association
`vix`	0.0010	0.0005	0.0010	n/s across all models
`post_crisis`	0.0299 ***	0.0096 *	0.0299 ***	FE controls country-level heterogeneity
R^2	0.7754	0.4070 (within)	0.7754	Balanced panel: N=160

6.3 Preferred Fixed Effects Model — CPI Detailed Results

Table 7: Fixed Effects Regression — CPI Model (Dependent Variable: ln CPI)

Variable	Coefficient	Robust Std. Err.	t-stat	p-value	Sig.
`oil_pct_change`	−0.000081	0.000126	−0.646	0.519	—
ln(NEER)	−0.3101	0.0985	−3.150	0.002	**
`policy_rate`	−0.0092	0.0028	−3.322	0.001	**
`vix`	0.0005	0.0007	0.710	0.479	—
`post_crisis`	0.0096	0.0046	2.095	0.038	*

*Entity fixed effects included. Arellano HAC robust SE. R^2 (within) = 0.407. N=160. ** $p < 0.01$, * $p < 0.05$.*

Within countries, the policy rate and the exchange rate are the most strongly significant correlates of CPI. A one-percentage-point higher policy rate is associated with a CPI level about 0.92% lower, a 1% stronger nominal effective exchange rate is associated with a CPI level about 0.31% lower, and the post-March-2026 period is associated with a CPI level about 0.96% higher. The oil price change itself and the VIX are not statistically significant, so the crisis effect is captured by the post-crisis dummy rather than by the monthly oil price change.

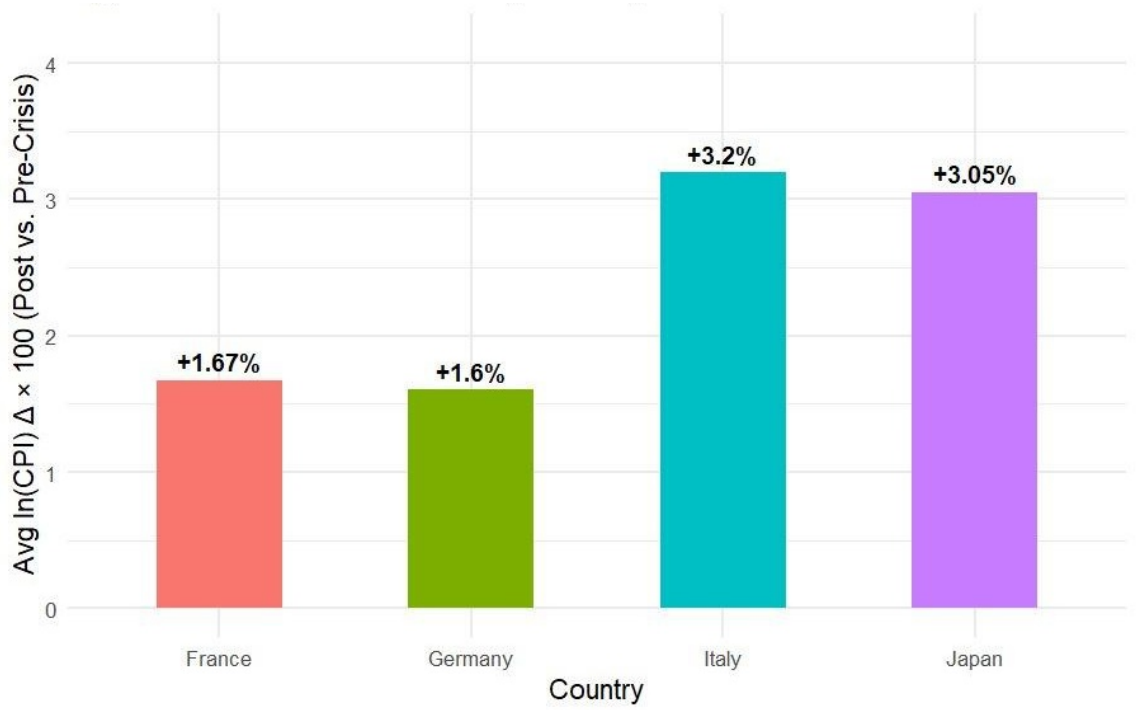


Figure 5: Post-crisis CPI lift by country. Japan (+3.05%) and Italy (+3.2%) register the largest lifts, consistent with limited monetary policy space and structural energy dependence respectively. Lifts are raw differences in average $\ln(CPI)$ *between the post- and pre-crisis windows, so they also reflect each series' pre-existing upward trend.*

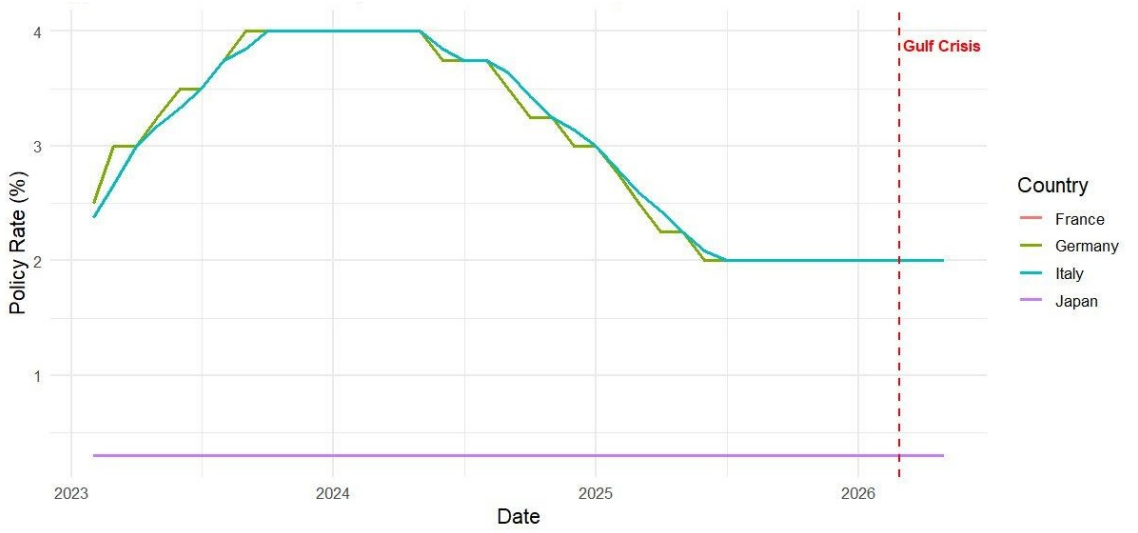


Figure 6: Central bank policy rates. Japan's near-zero rate (0.30%) is structurally distinct from the ECB's tightening-and-cutting cycle (peak 4.0% in 2023–2024).

6.4 IPI Model — Testing for Stagflationary Output Effects

Table 8: Fixed Effects Regression — IPI Model (Dependent Variable: ln IPI)

Variable	Coefficient	Robust Std. Err.	t-stat	p-value	Sig.
`oil_pct_change`	−0.000037	0.000126	−0.296	0.768	—
ln(NEER)	0.1274	0.0506	2.519	0.013	*
`policy_rate`	−0.0003	0.0014	−0.189	0.850	—
`vix`	0.0003	0.0006	0.403	0.687	—
`post_crisis`	−0.0009	0.0026	−0.322	0.748	—

*Entity fixed effects included. Arellano HAC robust SE. R^2 (within) = 0.063. N=160. * $p < 0.05$.*

The oil shock coefficient (−0.000037) and the post-crisis coefficient (−0.0009) have the negative sign expected for output, but neither is statistically distinguishable from zero. The VIX (0.0003, $p = 0.687$) and policy rate (−0.0003, $p = 0.850$) coefficients are also insignificant, and the within R^2 is only 0.063. The only significant coefficient is ln(NEER) (0.1274, $p = 0.013$). The opposite signs of the post-crisis effect in the CPI model (+) and the IPI model (−) are consistent in direction with a stagflationary narrative, but the IPI model provides no statistically significant evidence of an output effect over this short window.

7. Discussion and Policy Implications

7.1 Why Fixed Effects and Why Japan Matters

Japan's CPI mean of 111.51 versus the European average of ∼122.4 represents a structural level difference. In POLS this contaminates all slope estimates. In FE Japan's individual intercept captures this difference, and slope estimates reflect only within-country dynamics. This is why the policy rate coefficient becomes statistically significant in FE (−0.0092, $p = 0.001$) while it is insignificant in POLS (−0.0024), and why the ln(NEER) coefficient changes sign from +0.3660 in POLS to −0.3101 in FE.

7.2 Post-Crisis CPI Impact by Country

Figure 5 reveals heterogeneity within the Advanced Importers group. Japan (+3.05%) and Italy (+3.20%) show the largest post-crisis CPI lifts, while France (+1.67%) and Germany (+1.60%) show more modest increases. Japan's larger lift is consistent with its limited monetary policy space (0.30% rate). Italy's large lift reflects its structural energy import dependence and fiscal constraints. These figures are raw post-versus-pre differences and are not adjusted for the pre-existing upward trend in each series.

7.3 Heterogeneity Note

The project guidelines request a heterogeneity analysis comparing oil importers vs. exporters. Since our panel consists exclusively of Advanced Net-Importers this split analysis is not applicable within our group, as directed by the professor.

7.4 Policy Implications

The policy rate coefficient ($\beta = -0.0092$, $p = 0.001$) indicates that, within countries, a one-percentage-point higher policy rate is associated with a CPI level about 0.92% lower, which is consistent with the disinflationary role of the ECB's tightening cycle. This is an association rather than a causal estimate, since policy rates fell over most of the sample while CPI trended upward. Japan's near-zero rate environment leaves limited conventional policy space; exchange rate policy and forward guidance may be more relevant

tools.

8. Limitations and Extensions

Small N (4 countries). With only four panel units, the power of the Hausman test is limited, and Arellano robust standard errors rely on a large number of clusters. They may therefore be too small here, so p-values close to 5% (for example the post-crisis dummy, $p = 0.038$) should be read with caution.

Short post-crisis window. Only three post-crisis months (March–May 2026) are available, and Japan's final months are ARIMA forecasts rather than observed values (see Section 3.2). Japan's CPI is sourced from the IMF International Financial Statistics database (series JPN.CPI._T.IX.M, IMF IFS CPI 5.0.0).

Trending series and raw lifts. CPI enters in logs of levels with an upward trend, and the model has no time trend or time effects, so the post-crisis dummy and the policy rate may partly capture trends rather than crisis effects. The country lifts in Figure 5 are likewise raw differences.

Data variation. In the dataset, Germany's CPI appears constant from early 2025 and the IPI series for Germany, Italy, and Japan appear constant from early 2024 (Figures 2 and 4). This limits the within-country variation available to the estimators, and the IPI results in particular should be read as exploratory.

Potential extensions. Arellano-Bond GMM, local projections, interaction terms, country-specific trends or first differences.

9. Conclusion

This project applied panel data econometrics to assess the macroeconomic impact of the 2026 Gulf Crisis on CPI and IPI in four advanced net-importing economies using 160 monthly observations (February 2023–May 2026). The diagnostic sequence (BP: $\chi^2 = 423.00$; Hausman: $\chi^2 = 198.30$; DW: $F = 8.84$; Heteroskedasticity: $\chi^2 = 29.39$) selected Fixed Effects with Arellano HAC robust standard errors throughout.

The FE CPI model finds: (i) a post-crisis effect of $+0.96\%$ ($p = 0.038$), significant at the 5% level but based on only three post-crisis months; (ii) a one-percentage-point higher policy rate associated with a 0.92% lower CPI ($p = 0.001$); and (iii) a negative and significant $\ln(\text{NEER})$ coefficient, while the oil price change and the VIX are not significant. The FE IPI model finds no statistically significant effect of the oil shock, VIX, policy rate, or crisis dummy on industrial production (within $R^2 = 0.063$), so the stagflationary hypothesis receives directional but not statistical support. Japan's raw post-crisis CPI lift ($+3.05\%$) is among the largest in the panel, an illustration of the exposure of structurally low-rate economies to external inflationary shocks, although raw lifts also reflect pre-existing trends.

10. R Script Reference

The well-commented R script used to produce all results in this report is submitted alongside this document as `Panel_Project_Complete.R`. All results are fully reproducible by running the script with the accompanying `panel_data_final.csv` dataset.

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